

2026S1 PH1012: Physics A
Electromagnetism and Circuits
Electric Fields

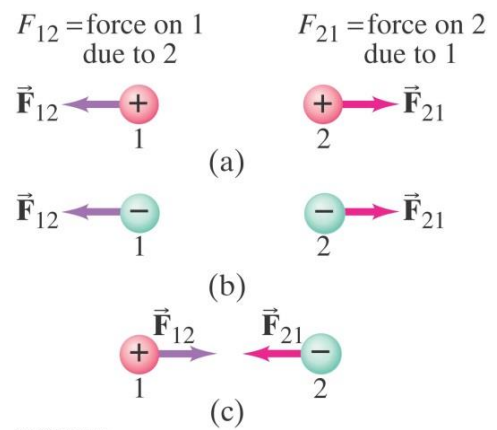
Dr Ho Shen Yong
Lecturer, School of Physical and Mathematical Sciences
Nanyang Technological University
Week 8
Serway Chap 22

"You can't depend on your judgment when your imagination is out of focus."

- Mark Twain

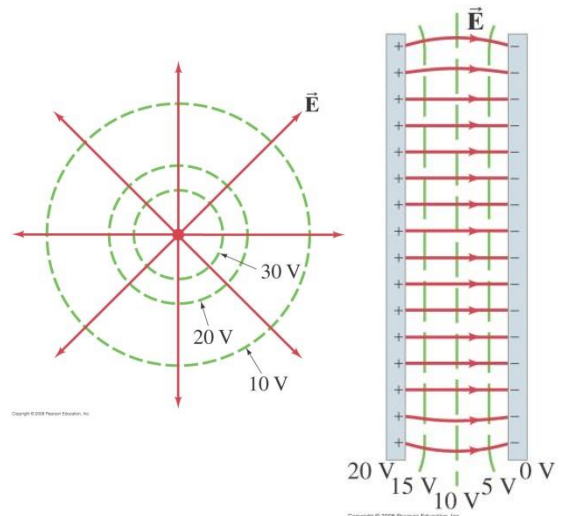
Itinerary

1. Electric forces and fields (for point charges and collection of point charges).
2. Magnetic Fields and Forces
3. Sources of Magnetic Fields
4. Electric fields, potentials and capacitance
5. Electric fields in circuits; resistors, capacitors and D.C circuits;
6. Kirchhoff's laws; RC circuits
7. Electromagnetic Induction
8. Gauss's law, Ampere's Law and Maxwell's Equation



Self-check – after completing the lessons (lecture and tutorial) for this topic, are you able to

1. cite examples of electrostatics and the effects of charge transfer
2. state Coulomb's law of electric forces and apply it solve problems in Physics
3. calculate the net force / field acting on a charge placed in a system of fixed charges or an external electric field
4. calculate electric field in problems and demonstrate understanding of the concept of electric force, field and flux



Introduction

In mechanics (dynamics), calculations often involve the physical property - mass of objects. In electricity and magnetism, the key physical property of objects that is of interest is electric charge. The subject of electricity and magnetism can be divided into three realms:

- 1) the charges that we study are at rest with respect to each other and to observers like us;
- 2) the charges are moving with a relatively constant average speed in the circuitry – “electricity”. Another phenomena that involves charges moving at constant velocity is the generation of magnetic fields.
- 3) the charges are accelerating – fields can be generated and energy radiated into space.

Gravitational force is important on scale of planets and our everyday life. However, electromagnetic interaction is the dominant force at the microscopic level down to the atomic scale. It is the force responsible for holding the electrons and the nuclei together in atoms and all forms of molecular bonding – thus responsible for all of chemical and biological phenomena. Looking at our present reliance on electrical appliances, we can also safely say that the understanding and application of electricity and magnetism is one of the cornerstone in the modernization of human civilization.

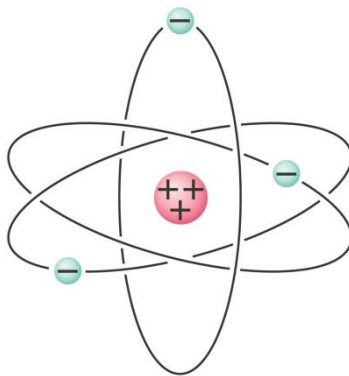
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The unit of charge: Coulomb (C)

The SI unit of charge is Coulomb (C) which is defined as the amount of charge that flows in 1 second when there is a steady current of 1 ampere. The electron is an elementary charge carrier and has a charge of $-e$ where

$$e = 1.60217733 \times 10^{-19} \text{C}.$$

The charge of a proton is $+e$. Thus, we see that one C is made up of many elementary charges.

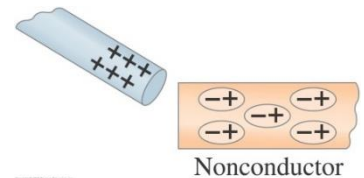
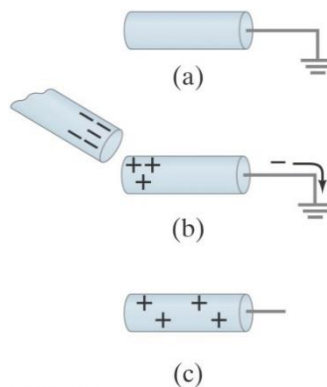
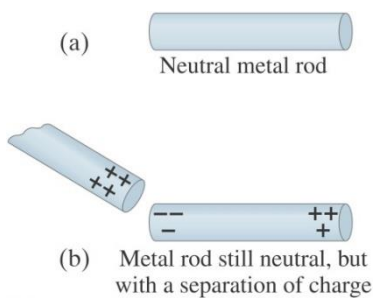
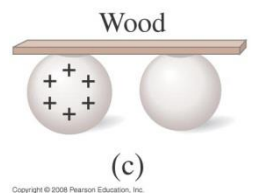
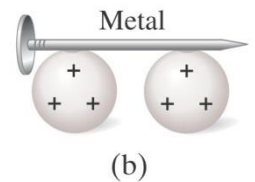
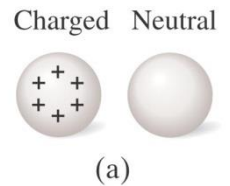
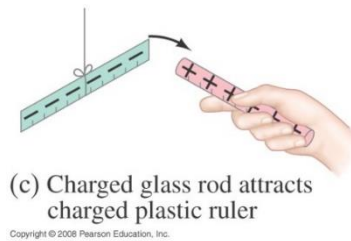
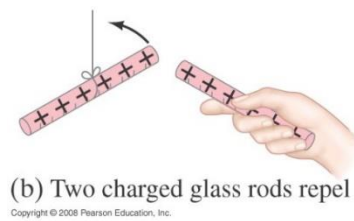
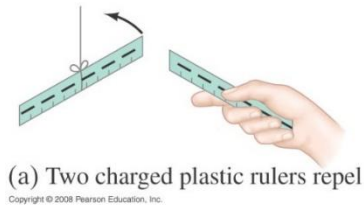
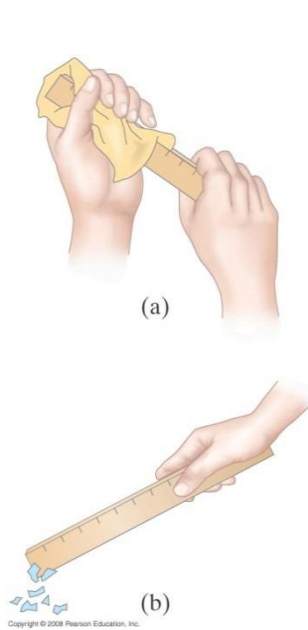


Giancoli pg 561 Fig 21.3:
Simple model of atom

Examples of Daily Occurrences and Applications

1. Lightning
2. Electrical igniters at gas stove
3. Charging of cellphones, camera, flash etc
4. Computer Keyboards
5. Capacitive Touchscreens

Aspects of Electrostatics



<https://www.youtube.com/watch?v=dwJ-MM7yu4E&t=3s>



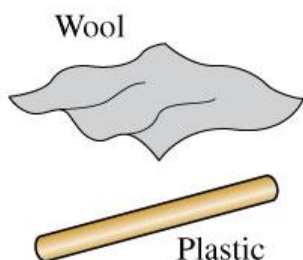
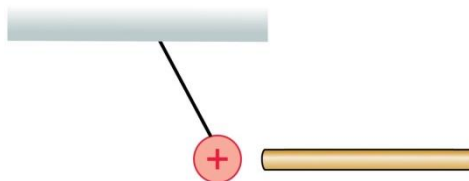
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Visualizing Charges (Knight)

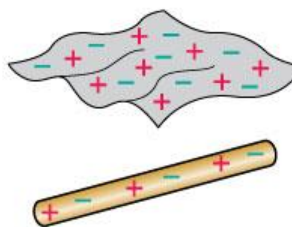
Pause to Ponder: A rod attracts a positively charged hanging ball.

The rod is

- A. Positive.
- B. Negative.
- C. Neutral.
- D. Either A or C.
- E. Either B or C.



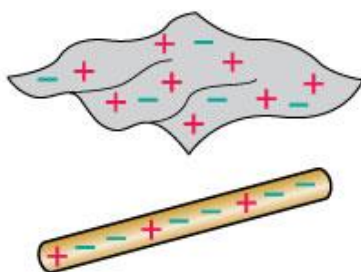
1. The rod and wool are initially neutral.



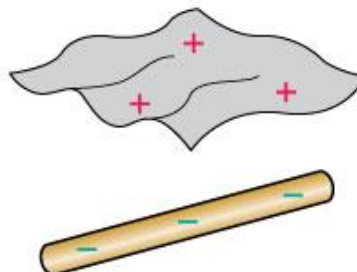
2. "Neutral" actually means that they each have equal amounts of positive and negative charge.



3. Now we rub the plastic and wool together. As we do so, negative charge moves from the wool to the rod.



4. This leaves the rod with extra negative charge. The wool is left with more positive charge than negative.

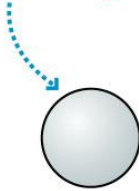


5. In a charge diagram, we draw only the *excess* charge. This shows how the rod and wool acquire their charge.

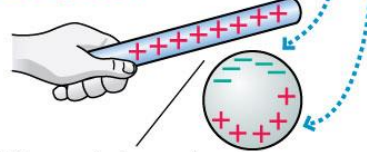
Charge Polarization (Knight)

Charge polarization is the slight separation of the positive and negative charges in a neutral object when a charged object is brought near.

The neutral sphere contains equal amounts of positive and negative charge.

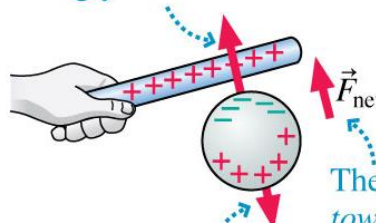


Negative charge is attracted to the positive rod. This leaves behind positive charge on the other side of the sphere.



The rod doesn't touch the sphere.

The negative charge on the sphere is close to the rod, so it is strongly attracted to the rod.



The *net* force is toward the rod.

The positive charge on the sphere is far from the rod, so it is weakly repelled by the rod.

The **polarization force** arises because the charges in a neutral object are slightly separated, *not* because the objects are oppositely charged.

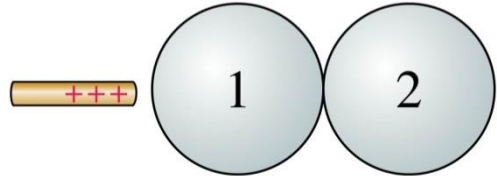
The polarization force between a charged object and a neutral one is always attractive.

Pause to Ponder: Metal spheres 1 and 2 are touching. Both are initially neutral.

- The charged rod is brought near.
- The charged rod is then removed.
- The spheres are separated.

Afterward, the charges on the sphere are:

- Q_1 is + and Q_2 is +
- Q_1 is + and Q_2 is -
- Q_1 is - and Q_2 is +
- Q_1 is - and Q_2 is -
- Q_1 is 0 and Q_2 is 0

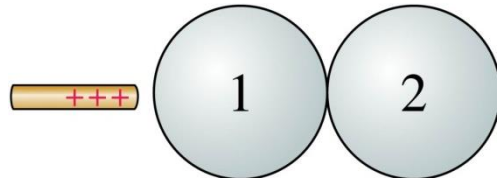


Pause to Ponder: Metal spheres 1 and 2 are touching. Both are initially neutral.

- The charged rod is brought near.
- The spheres are separated.
- The charged rod is then removed.

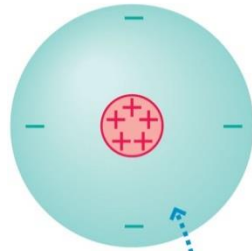
Afterward, the charges on the sphere are:

- Q_1 is + and Q_2 is +
- Q_1 is + and Q_2 is -
- Q_1 is - and Q_2 is +
- Q_1 is - and Q_2 is -
- Q_1 is 0 and Q_2 is 0



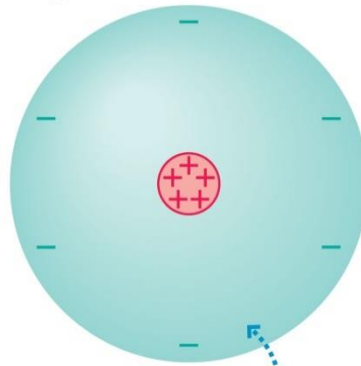
Atomic view of charging (Knight)

Positive ion



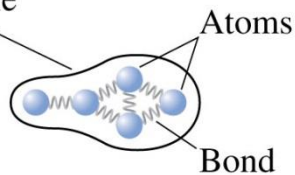
The atom has lost one electron, giving it a net positive charge.

Negative ion

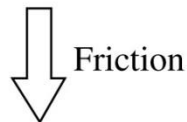


The atom has gained one electron, giving it a net negative charge.

Electrically neutral molecule

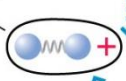


Molecular ions can be created when one of the bonds in a large molecule is broken.

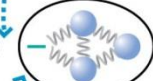


These bonds were broken by friction.

Positive molecular ion



Negative molecular ion



This half of the molecule lost an electron as the bond broke.

This half of the molecule gained an extra electron as the bond broke.

Wimshurst Machine and Van der Graaff generator



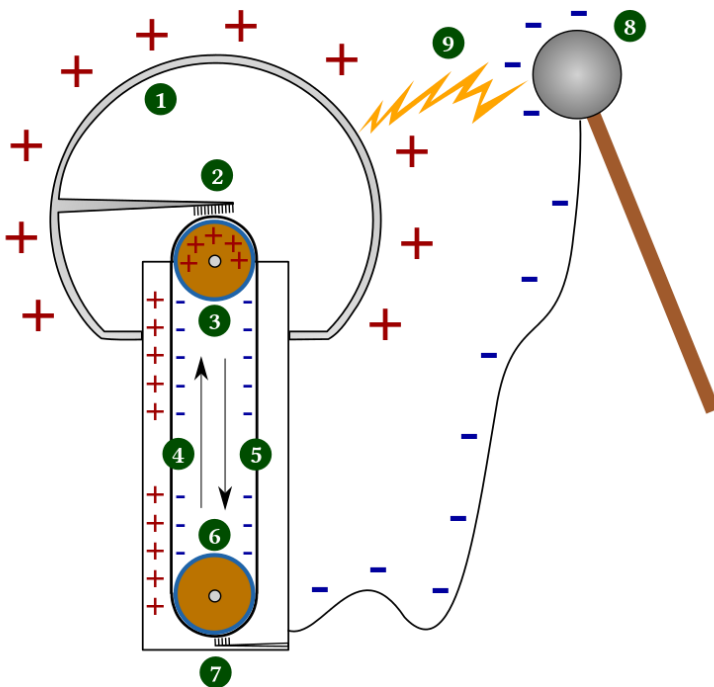
<https://commons.wikimedia.org/wiki/File:Wimshurst.jpg>

How it works: See Youtube

[MIT Physics Demo – The Wimshurst Machine](#)

[Hand-Cranked Laser – TEA Laser Powered by Wimshurst Machine](#)

Van de Graaff Generator



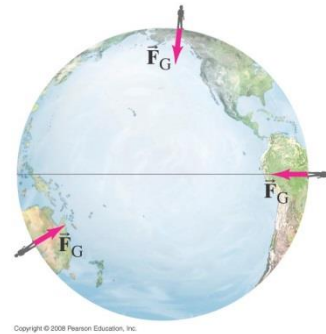
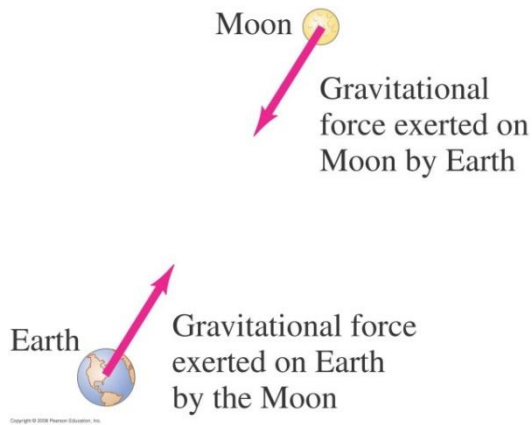
- | | |
|---|---|
| 1. hollow metal sphere | 6. lower roller (metal) |
| 2. upper electrode | 7. lower electrode (ground) |
| 3. upper roller (for example an acrylic glass) | 8. spherical device with negative charges |
| 4. side of the belt with positive charges | 9. spark produced by the difference of potentials |
| 5. opposite side of belt, with negative charges | |



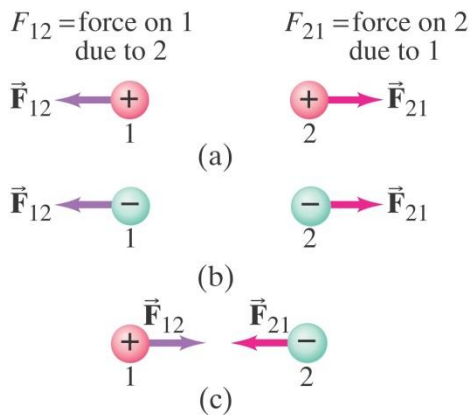
https://en.wikipedia.org/wiki/Van_de_Graaff_generator

Newton's Law of Gravitation and Coulomb's Law

Newton's Law of Gravitation



Coulomb's Law on Electrostatic forces



Coulomb's Law:

Gauss's Law which was discussed for gravitational fields is also similarly applicable to electric fields.

Coulomb's law (discovered in 1785)

The magnitude of the force (F) between two electrically charged bodies, which are small compared with their separation (r), is inversely proportional to r^2 and proportional to the product of their charges (Q_1 and Q_2):

$$F = k \frac{Q_1 Q_2}{r^2}$$

where the constant of proportionality k , in vacuum, expressed in SI units is given by

$$k = \frac{1}{4\pi\epsilon}$$

where ϵ is the permittivity. It is dependent on the medium in which the charges are in. For example, the permittivity of water at room temperature is about 80 times the permittivity of free space (vacuum) and thus the attractive forces between Na^+ and Cl^- is much weaker in water. The permittivity of free space (vacuum)

$$\epsilon_0 = 8.85418781762 \times 10^{-12} \frac{\text{C}^2}{\text{N} \cdot \text{m}^2}.$$

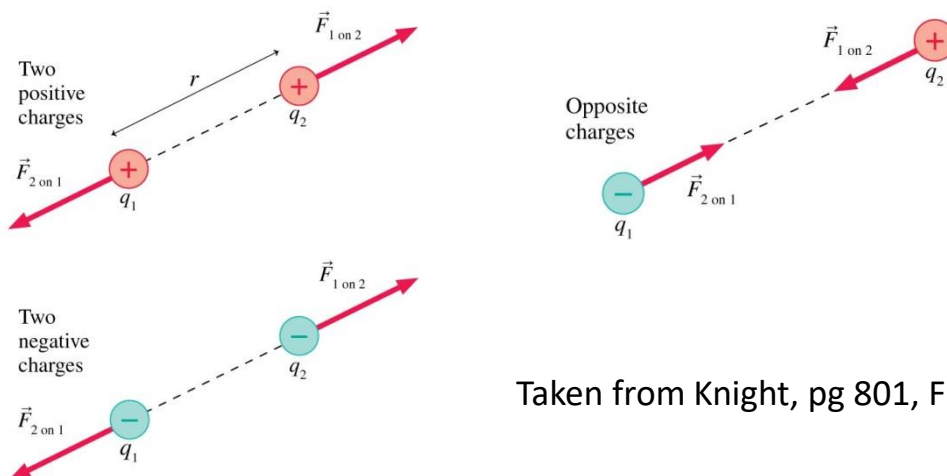
For the purpose of simple computation in vacuum, we just use

$$k = \frac{1}{4\pi\epsilon_0} = 8.99 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2$$

Thus, Coulomb's law can be written as

$$F = \frac{Q_1 Q_2}{4\pi\epsilon_0 r^2}.$$

FIGURE 26.17 Attractive and repulsive forces between charges.



Taken from Knight, pg 801, Fig 26.17

Gravitational Force Vs Electric Force

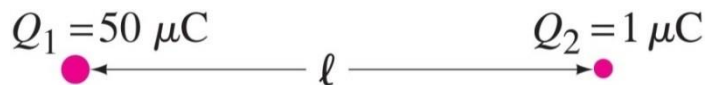
Calculate the gravitational force of attraction between two 1 kg mass, with separation 0.5 m.

Calculate the electric force of between two objects with 1 C charge, with separation 0.5 m.

Example Giancoli page 565, Conceptual Example 21-1: (modified)

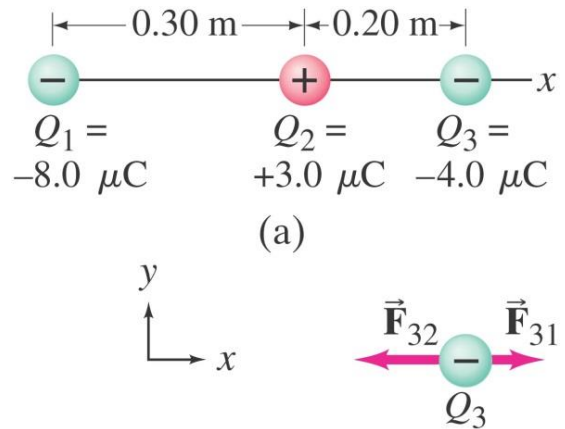
Which charge exerts the greater force?

Two positive point charges, $Q_1 = 50 \mu\text{C}$ and $Q_2 = 1 \mu\text{C}$, are separated by a distance $l = 0.50 \text{ m}$. Calculate the force that Q_1 exerts on Q_2 . What about the force that Q_2 exerts on Q_1 , will it be larger?



Giancoli pg 566 Example 21-2: Three charges in a line

Three charged particles are arranged in a line, as shown. Calculate the net electrostatic force on particle 3 (the $-4.0 \mu\text{C}$ on the right) due to the other two charges.



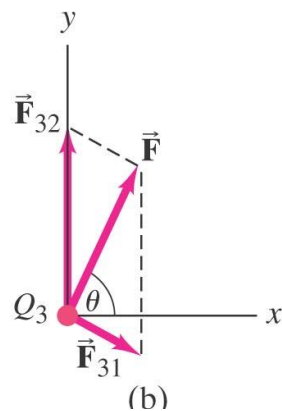
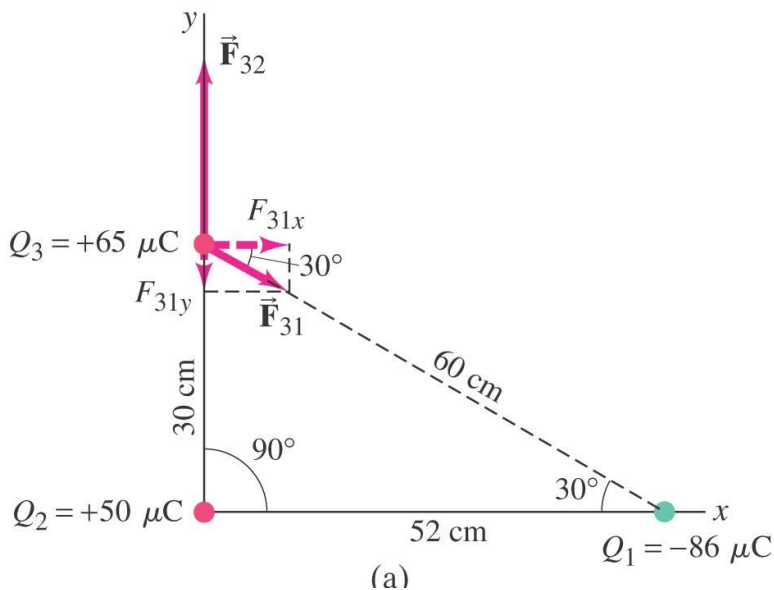
$$|\vec{F}_{32}| =$$

$$|\vec{F}_{31}| =$$

$$\vec{F} =$$

Giancoli pg 566 Example 21-3: Electric force using vector components.

Calculate the net electrostatic force on charge Q_3 shown in the figure due to the charges Q_1 and Q_2 .



$$|\vec{F}_{32}| =$$

$$|\vec{F}_{31}| =$$

$$F_x =$$

$$F_y =$$

$$|\vec{F}| =$$

$$\theta =$$

Electric field of a point charge

The electric field exist in the region of space around a charged object, the source charge Q . To study the properties of the electric field, a very small positive test charge q_0 is placed in this region to probe the electric field without distorting it. The test charge q_0 will experience a force $\vec{F}_e$ due to the source charge. The electric field $\vec{E}$ of the charge Q at a point is defined as

$$\vec{E} \equiv \lim_{q_0 \rightarrow 0} \frac{\vec{F}_e}{q_0}$$

Remarks:

General

1. $\vec{E}$ has SI units N/C.
2. The direction of $\vec{E}$ is the direction of force the positive test charge experiences.
3. The electric field is a property of the charge Q , independent of the test charge.
4. We can write the force $\vec{F}_e$ acting on any charge q in an electric field $\vec{E}$ as
$$\vec{F}_e = q\vec{E}.$$

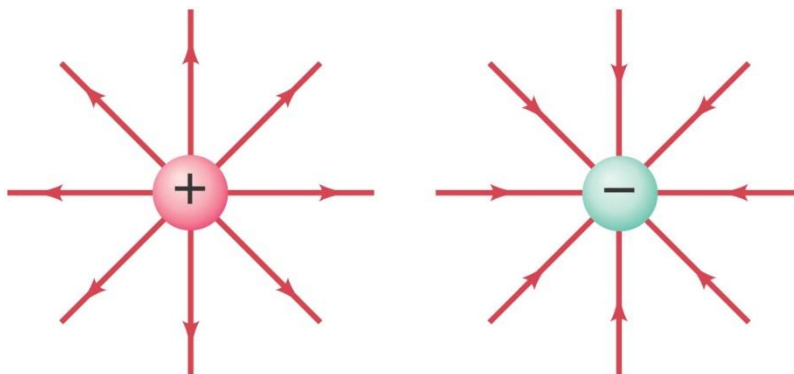
This is true for any electric field.

5. **For a point charge Q ,**

$$\vec{E} = \frac{\vec{F}_e}{q_0} = \frac{Q}{4\pi\epsilon_0 r^2} \hat{r}$$

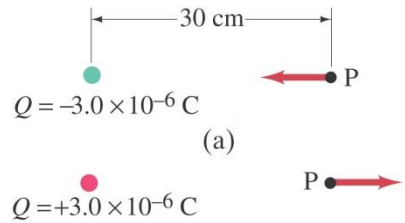
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The electric field can be represented by field lines. These lines start on a positive charge and end on a negative charge.



Giancoli page 569, Example 21-6: Electric field of a single point charge.

Calculate the magnitude and direction of the electric field at a point P which is 30 cm to the right of a point charge $Q = -3.0 \times 10^{-6} \text{ C}$.



What will be the electric field strength at another point

- 15 cm away from Q (+ve)?
- 60 cm away from Q (+ve)?

Point Charges

Resultant electric fields of multiple charges in general

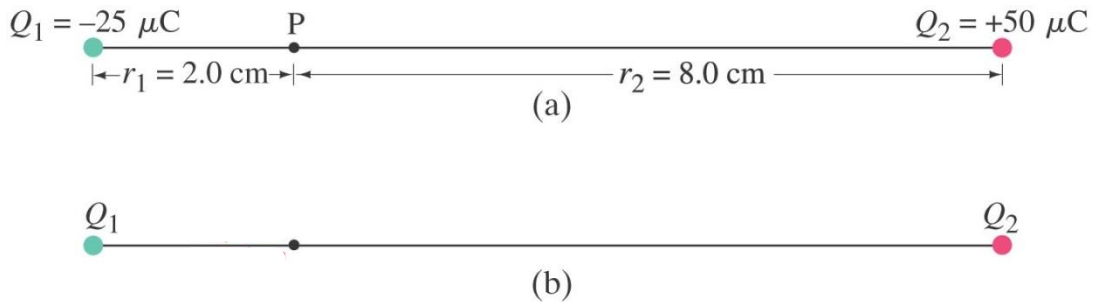
General

To find $\vec{E}$ for a group of N point charges, the procedure is as follows:

- 1) Calculate $\vec{E}_i$ due to charge i at the given point as if it were the only charge present.
- 2) Add these separately calculated fields vectorially to find the resultant field $\vec{E}$ at the point. In equation form,
$$\vec{E} = \vec{E}_1 + \vec{E}_2 + \vec{E}_3 + \dots$$
- 3) The equation is an example of the application of *superposition*. However, this approach may fail when the magnitudes of the fields are extremely large (perhaps causing polarization of non-elementary source charges) but in the present course, it is always valid.

Giancoli pg 570 Example 21-7: E at a point between two charges.

Two point charges are separated by a distance of 10.0 cm. One has a charge of $-25\ \mu\text{C}$ and the other $+50\ \mu\text{C}$. (a) Determine the direction and magnitude of the electric field at a point P between the two charges that is 2.0 cm from the negative charge. (b) If an electron (mass = $9.11 \times 10^{-31}\ \text{kg}$) is placed at rest at P and then released, what will be its initial acceleration (direction and magnitude)?

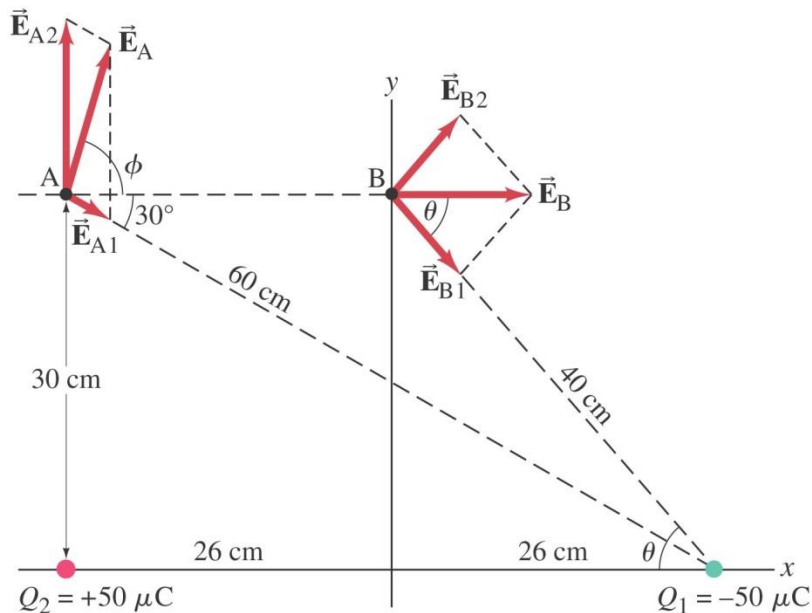
**Point Charges****Problem solving in electrostatics: electric forces and electric fields**

1. Draw a diagram; show all charges, with signs, and electric fields and forces with directions.
2. Calculate forces using Coulomb's law.
3. Add forces vectorially to get result.
4. Check your answer!

General

Giancoli page 570, Example 21-8: Electric field of two point charges.

Calculate the total electric field (a) at point A and (b) at point B in the figure due to both charges, Q_1 and Q_2 .



$$|\vec{E}_{A2}| = \frac{kQ_2}{r^2} = \frac{\left(9.0 \times 10^9 \text{ N} \frac{\text{m}^2}{\text{C}^2}\right) (50 \times 10^{-6} \text{ C})}{(60 \text{ cm})^2} =$$

$$|\vec{E}_{A1}| = \frac{kQ_1}{r^2} = \frac{\left(9.0 \times 10^9 \text{ N} \frac{\text{m}^2}{\text{C}^2}\right) |(-50 \times 10^{-6} \text{ C})|}{(30 \text{ cm})^2} =$$

$$E_{Ax} =$$

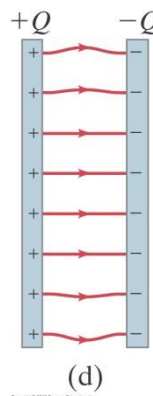
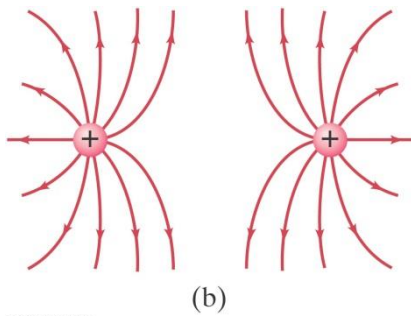
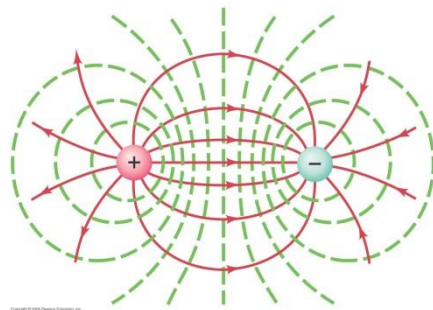
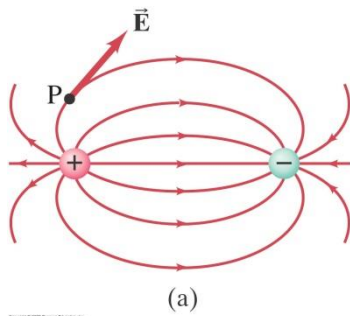
$$E_{Ay} =$$

$$|E_A| =$$

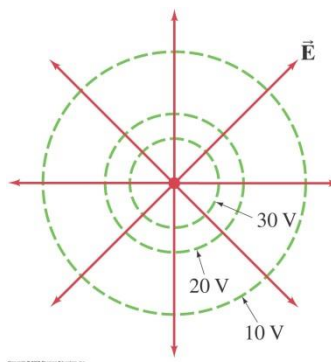
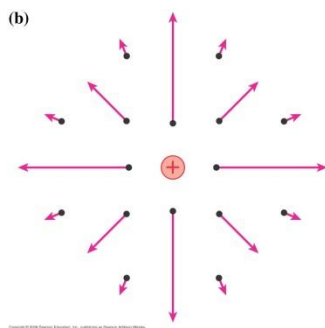
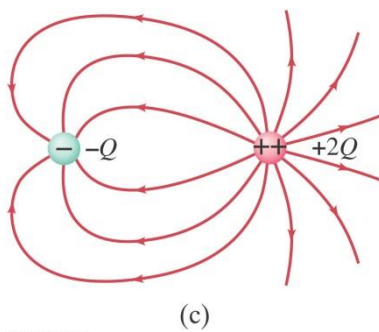
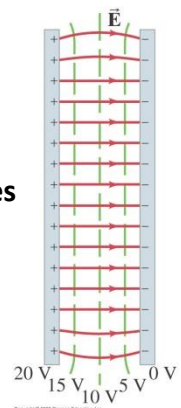
$$\theta =$$

Solve for E_B yourself

Electric fields of various systems of charges



Parallel Plates



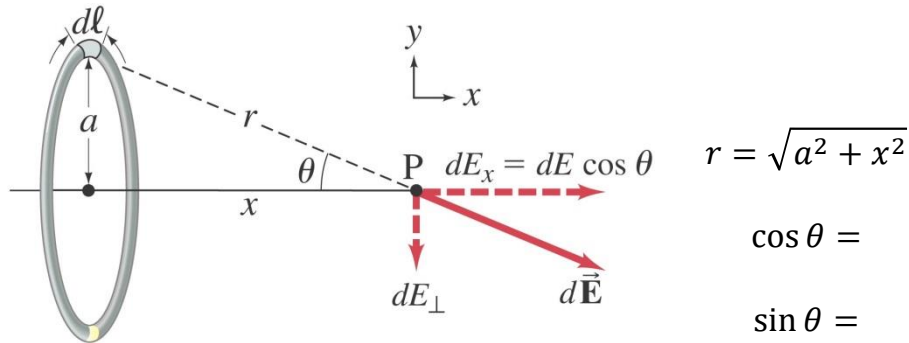
<https://www.youtube.com/watch?v=pOPQBlnBcM&t=3s>



<https://phet.colorado.edu/en/simulations/charges-and-fields>

Giancoli pg 572 Example 21-9: A ring of charge. [Non-Examinable]

A thin, ring-shaped object of radius a holds a total charge $+Q$ distributed uniformly around it. Determine the electric field at a point P on its axis, a distance x from the center. Let λ be the charge per unit length (C/m).



A small segment of length dl has charge $dq = \lambda dl$ where $\lambda = \frac{Q}{2\pi a}$

The contribution of charge dq to the magnitude of electric field at P is

$$dE = \frac{k dq}{r^2} = \frac{k \lambda dl}{r^2}$$

$d\vec{E}$ has two components - $dE_{\perp}$ and dE_x

Summing up all the components perpendicular ($\perp$) to x direction, we get

$$E_{\perp} = 0$$

Summing up all the components along the x direction, we get

$$\begin{aligned} E_x &= \int dE_x = \int dE \cos \theta = k \lambda \int \frac{dl}{r^2} \cos \theta \\ &= \frac{k \lambda}{r^2} \cos \theta \int_0^{2\pi a} dl = \end{aligned}$$

Check for reasonableness – when $x \gg a$, $E =$

Comparison of Gravitational, Electric and Magnetic Fields

Property			
Forces			
Fields			

Flux

Can you cause moving charges / wire carrying current to deflect?

Dr Ho Shen Yong

Lecturer, School of Physical and Mathematical Sciences

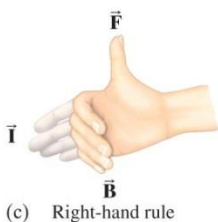
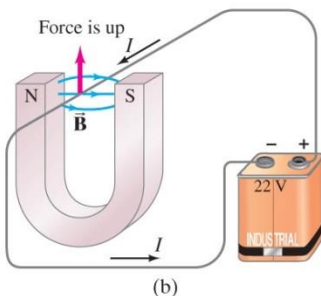
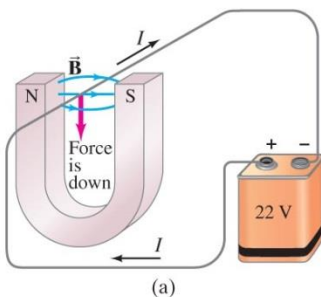
Nanyang Technological University

Week 9

Serway Chap 28

"Nikola Tesla is the true unsung prophet of the electronic age; without whom our radio, auto ignition, telephone, alternating current power generation and transmission, radio and television would all have been impossible."

Ben Johnston in the "Introduction" to My Inventions :
The Autobiography of Nikola Tesla (1983))



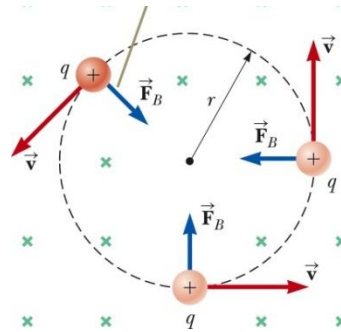
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$$\vec{F}_B = I\vec{L} \times \vec{B}$$

$$|\vec{F}_B| = ILB \sin \theta$$

Self-check – after completing the lessons (lectures and tutorials) for this topic, are you able to

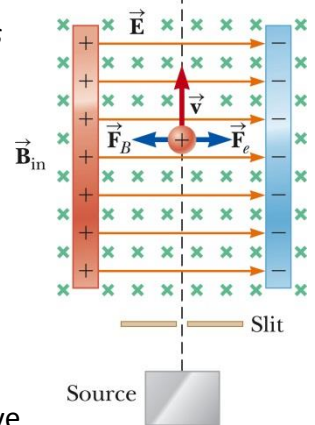
1. determine magnetic forces acting on a moving free charge or a current carrying conductor and analyze the effects on the motion
2. determine magnetic fields in associate with current-carrying conductors
3. integrate the concepts in 1 – 2 above to solve problems in electromagnetism



$$\vec{F}_B = q\vec{v} \times \vec{B}$$

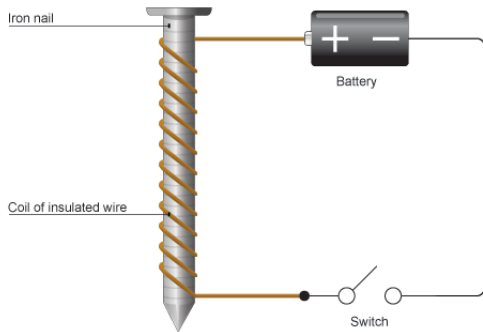
$$|\vec{F}_B| = q|\vec{v}||\vec{B}| \sin \theta$$

Serway pg 834



$$\vec{F} = q\vec{E} + q\vec{v} \times \vec{B}$$

Some primary science lessons



More turns per unit length
round nail =>

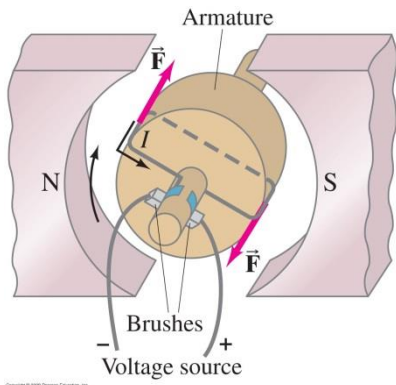
Stronger battery =>

Reversing sign of battery=>

Removing nail =>



<https://www.youtube.com/watch?v=4zvvAjj18Dk&t=4s>



Giancoli 27-23 DC Motor

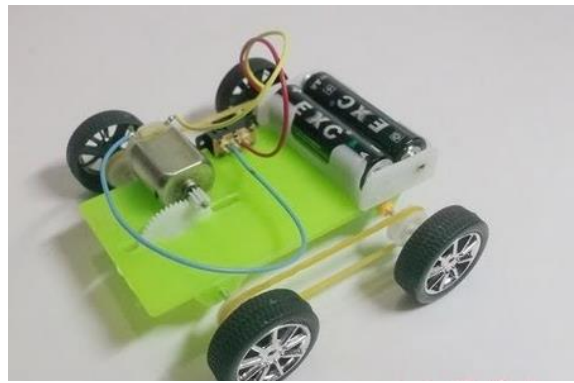
More turns of wire on armature=>

Stronger battery =>

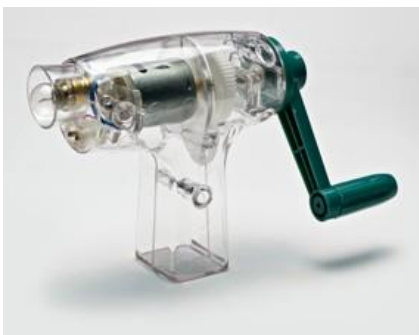
Stronger magnet=>

Longer wires=>

Reversing sign of battery=>



https://www.youtube.com/watch?v=6NQAwt31_Jo&t=2s



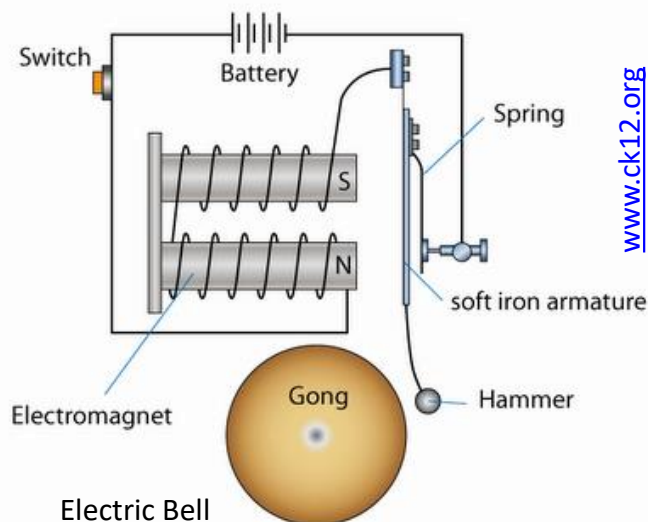
Turning handle faster=>

Stronger magnet=>

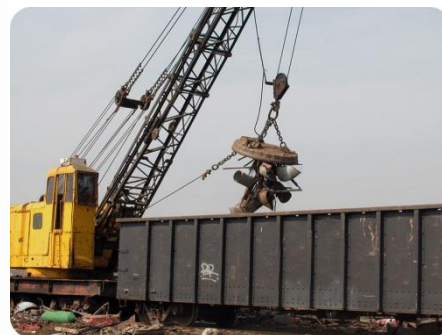
Reversing direction of turn=>

Some more applications of Electromagnetism (Secondary School / JC)

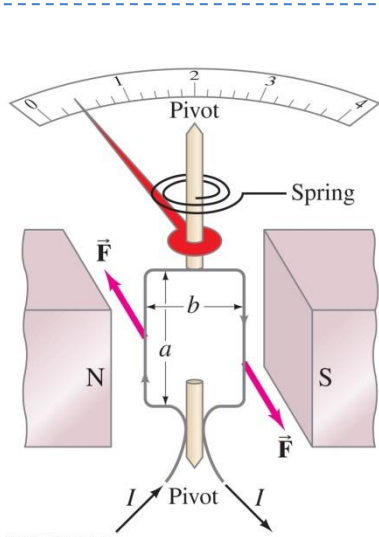
www.superteachertools.net



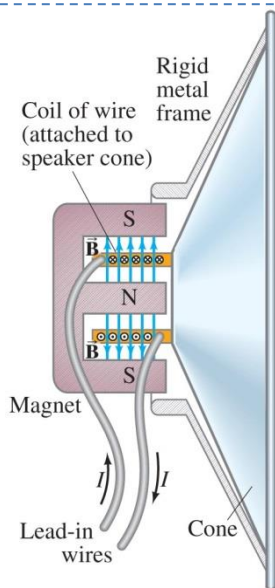
www.ck12.org



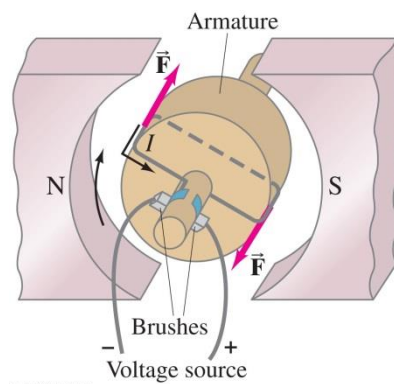
Electromagnet at junk yard



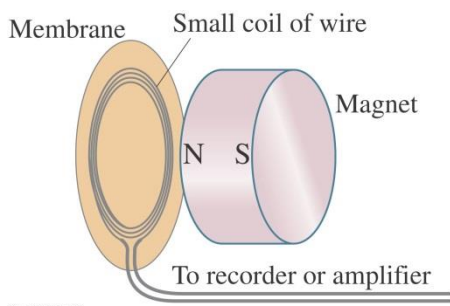
Giancoli 27-27 Galvanometer



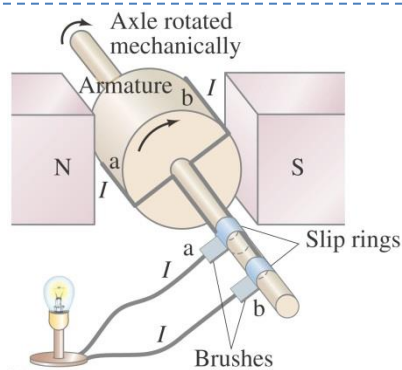
Giancoli 27-26 Speaker



Giancoli 27-23 DC Motor



Giancoli 29-28 Microphone



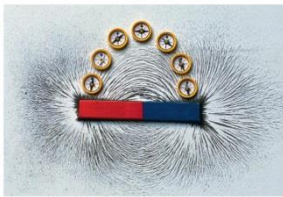
Giancoli 29-15 AC Generator

Introduction

We have studied two of the three realms of the subject of electricity and magnetism. We dealt with the static electrical charges in electric fields and charges moving with relatively constant velocities in current electricity. In the next three lectures, we will be studying magnetic fields generated by moving charges and how accelerating charges can generate fields.

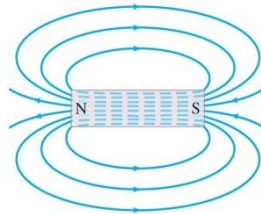
The phenomena of magnetism pervades all scales in nature. On the planetary scale, within the Earth is a giant magnet - it has a hot iron core which carries a current when it moves, producing a protective magnetic field around the Earth. The Earth's magnetic field shields us from the solar wind, a stream of (harmful) energetic charged particles from the Sun. As we will learn, these charged particles are deflected by Earth's magnetic field.

At the microscopic level, we speak of the magnetic moments of quantum particles. They play an important role in determining the behaviour of atoms and molecules in magnetic fields. The phenomena of magnetism is also exploited in many technologies – such as data storage, mobile phones, speakers, microphones, Magnetic Resonance Imaging (MRI), propulsion using magneto-hydrodynamics (MHD), magnetic levitation train...



(a)

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(b)

Giancoli 27-4

Itinerary

Parallel in Electric (E) Fields

Charges and E-Fields

Coulomb's Law (E fields)

Gauss's Law for E fields

Coulomb's Law (Forces)



Lecture

Moving charges / Current
=> Magnetic (B) Fields

Biot-Savart's Law

Ampere's Law

Gauss's Law in B Fields

Moving charges / Current
in B Fields
=> Deflection force

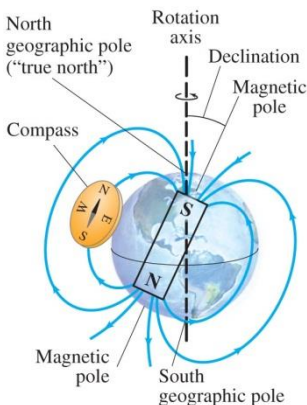
Motion (conductor) in B Fields
=> generate electricity (e.m.f.)

About magnetic fields (B)

The unit for magnetic field are those that we deduced from the interaction between two parallel currents

$$B = \frac{\text{Force}}{[\text{Current} \cdot \text{Length}]} = \frac{N}{A \cdot m}$$

The unit magnetic field is telsa (T) in honour of Nikola Tesla (1856 – 1943). The tesla is a large unit of magnetic field. A smaller unit for the magnetic field is the gauss (G): 1T = 10 000 G. The magnetic field is a vector and the strength of a magnetic field is indicated by the flux density (lines per unit area).

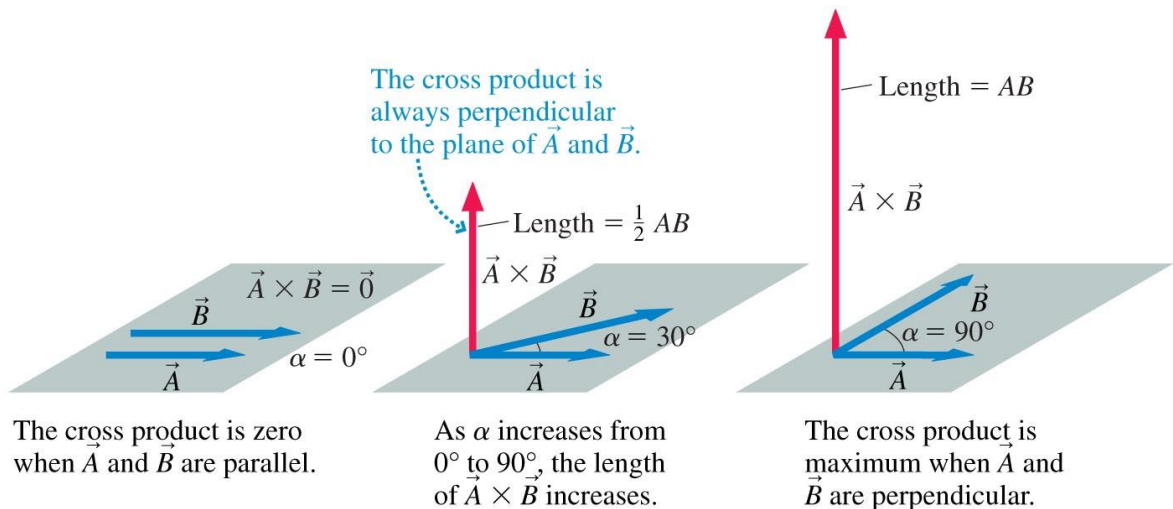
Source	Magnetic Field Strength (T)
Earth (near surface)  Giancoli pg 709 Fig. 27.5	About 5×10^{-5}  https://www.youtube.com/watch?v=yEYynVC4L0&t=3s
Small bar magnet	10^{-2}
Human Exposure Limit (d.c for minutes)	0.2
Brain's a rhythm currents	10^{-15}
Colour TV (CRT)	10^{-4}
Transmission line	About 4×10^{-5}
Beam focusing research magnet or electric motor	1
Superconducting research magnets	10
1 m long wire carrying 1A	2×10^{-7}

Cross (Vector) Product (Recap)

The treatment of electric (E) (or gravitational) fields is much more straight forward compared to magnetic fields. This is because the direction of the electric forces acting on charges in E-fields are parallel to the fields: $\vec{F}_e = q\vec{E}$. On the contrary, for example, magnetic forces are perpendicular to magnetic fields and we need Fleming's left hand rule to determine the directions of the force. The most appropriate mathematical tool to use in the description of phenomena associated with magnetic fields is the cross (vector) product:

$$\vec{A} \times \vec{B} = |\vec{A}||\vec{B}| \sin \alpha \hat{n}$$

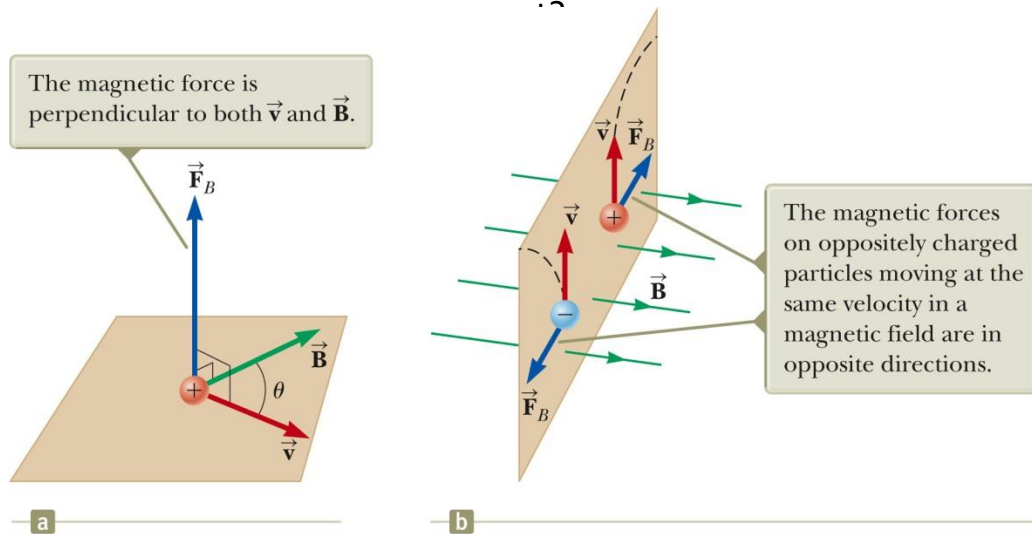
where $\hat{n}$ is the unit vector perpendicular to both $\vec{A}$ and $\vec{B}$. The orientation of $\hat{n}$ follows the right hand grip.



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[Knight pg 369; Fig 12.51]

How to produce a deflection force on moving charges / wire carrying?



[Serway pg 833; Fig 29.4]

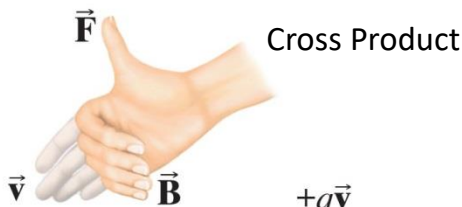
The deflection force $\vec{F}_B$ (N) acting on a charge q (C) moving with velocity $\vec{v}$ (m/s) in a magnetic field $\vec{B}$ (T) is given by

$$\vec{F}_B = q\vec{v} \times \vec{B}$$

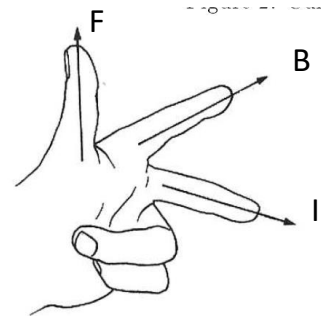
$$|\vec{F}_B| = q|\vec{v}||\vec{B}|\sin\theta$$

Here, the force $\vec{F}$ is always perpendicular to $\vec{v}$ so the magnitude of the velocity does not change. To determine direction, use either

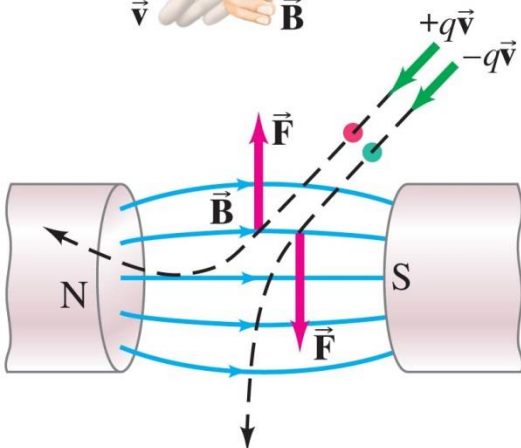
Right-hand rule



or



Fleming's Left Hand Rule



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<https://www.youtube.com/watch?v=ZS8hpX8nlfA>

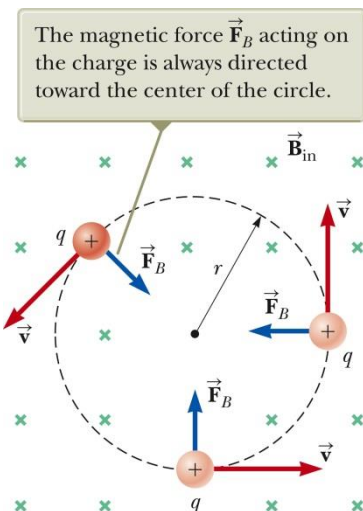
Exercise: (Giancoli Prob 27.13) Determine the magnitude and direction of the force on an electron travelling at 8.75 m/s horizontally to the East in a vertically upward magnetic field of strength 0.45 T .



<https://www.youtube.com/watch?v=RqSode4HZrE&t=4s>

Deflection of charges moving in a magnetic field

In fact, in a uniform magnetic field, the charge will move in a circular arc and $\vec{F}_B$ will always point towards the center of the circle.



$$F_B = qvB = \frac{mv^2}{r} (= mr\omega^2)$$

The radius of the circle is

$$r = \frac{mv}{qB}$$

and the angular velocity is

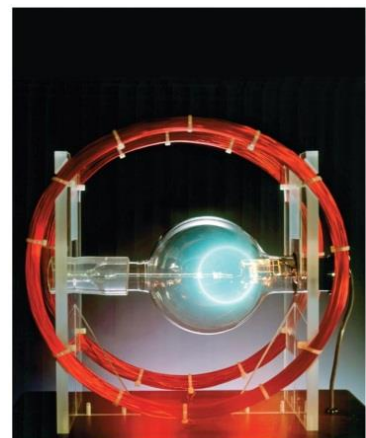
$$\omega = \frac{qB}{m}.$$

Note that now we can find out the mass-charge ratio: B and r can be found easily; we can use the velocity selector (see later) to find v .

$$\frac{m}{q} = \frac{rB}{v}$$

Giancoli pg 715, Example 27-7

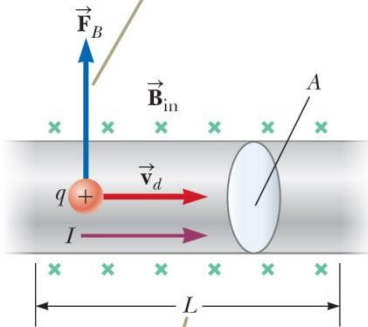
An electron travels at $2.0 \times 10^7 \text{ m/s}$ in a plane perpendicular to a uniform 0.010-T magnetic field. Describe its path quantitatively.



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Magnetic force on wire carrying current [Serway pg 841; Chap 29.4, Giancoli pg 710]

The average magnetic force exerted on a charge moving in the wire is $q\vec{v}_d \times \vec{B}$.



The magnetic force on the wire segment of length L is $I\vec{L} \times \vec{B}$.

The deflection force $\vec{F}_B$ (N) acting on a single charge q (C) moving with velocity $\vec{v}$ (m/s) in a magnetic field $\vec{B}$ (T) is given by $\vec{F}_B = q\vec{v} \times \vec{B}$.

For a section of a wire of length L immersed in a uniform magnetic field $\vec{B}$, the total force acting on all the charge carriers moving at drift velocity v_d is given by F = force on one charge carrier $\times$ total number of charge carriers in L

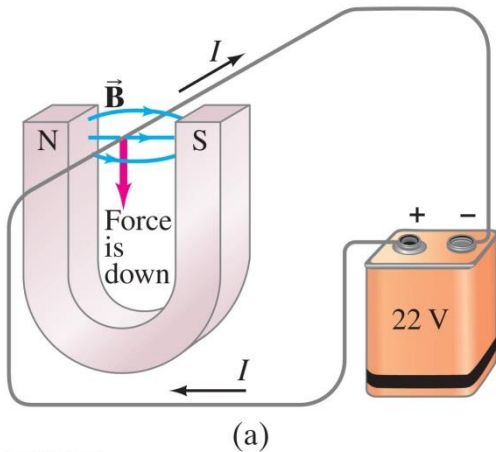
$$\vec{F}_B = (q\vec{v}_d \times \vec{B})n_eLA$$

$$= (n_eq\vec{v}_dAL \times \vec{B})$$

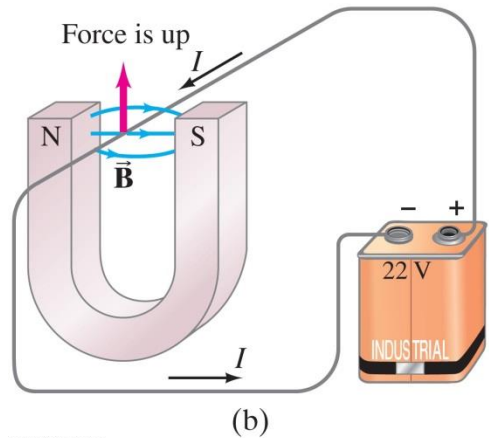
$$\vec{F}_B = I\vec{L} \times \vec{B}$$

$$|\vec{F}_B| = ILB \sin \theta$$

since current $I = n_eqv_dA$ and $\vec{L}$ points in the direction of current. $\vec{F}_B$ is perpendicular to both $\vec{L}$ and $\vec{B}$.

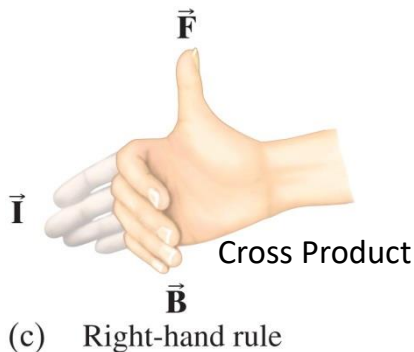


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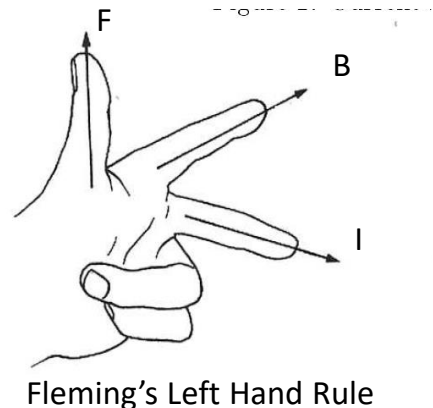
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Use Either



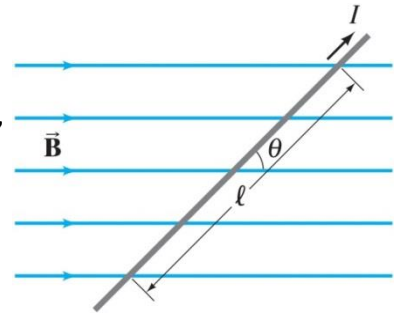
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or



Giancoli pg 712 Example 27.1

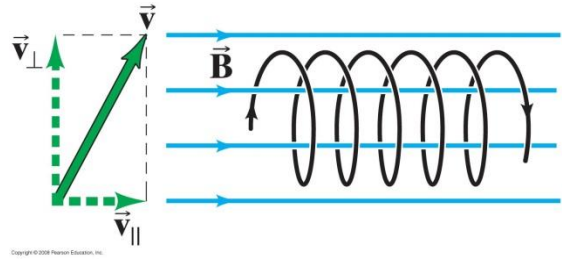
A wire carrying a 30-A current has a length $l = 12$ cm between the pole faces of a magnet at an angle $\theta = 60^\circ$, as shown. The magnetic field is approximately uniform at 0.90 T. We ignore the field beyond the pole pieces. What is the magnitude of the force on the wire?



<https://www.youtube.com/watch?v=fHuhkNosSrg&t=5s>

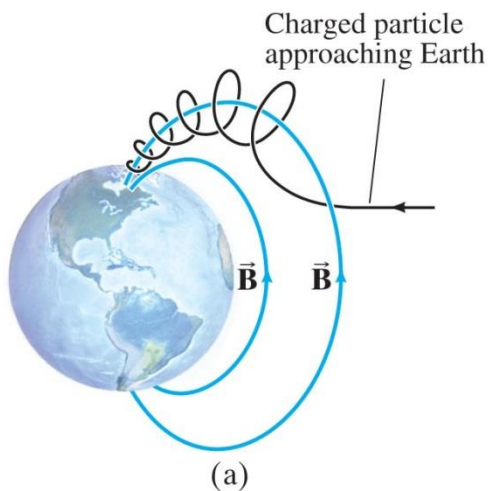
Magnetic (deflection) forces in action (Giancoli 27.4)

For $F = Bqv \sin \theta$, we have always considered on cases whereby the angle θ between $\vec{B}$ and $\vec{v}$ is 90° . What is the path of a charged particle in a uniform magnetic field if $\theta \neq 90^\circ$ as shown?



For $v_\perp$ to $\vec{B}$,

For $v_\parallel$ to $\vec{B}$,

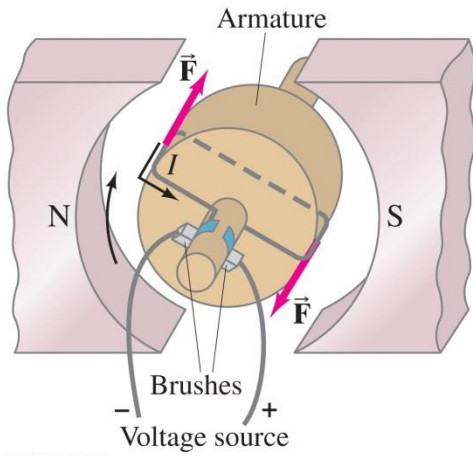


What happens when a stream of high speed charged particles from the Sun approach Earth? Will we be harmed?

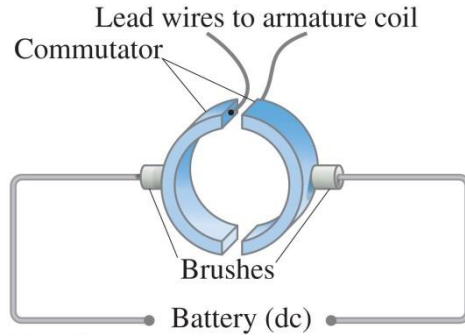
For $v_\perp$ to $\vec{B}$,

For $v_\parallel$ to $\vec{B}$,

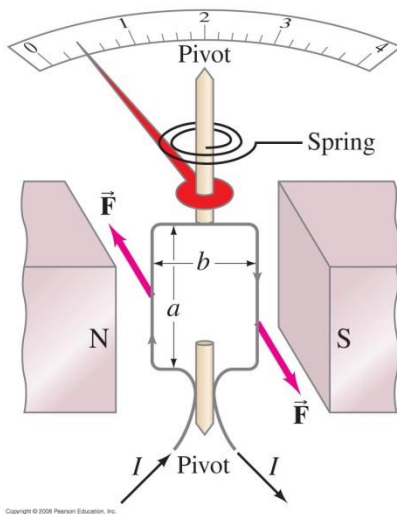
Magnetic (deflection) forces in action (Giancoli 27.6)



An electric motor uses the torque on a current loop in a magnetic field to convert magnetic energy into kinetic energy.



The commutator ring ensures that the motor always turn in the same one direction.



A galvanometer takes advantage of the torque τ on a current loop to measure current; the spring constant is calibrated so the scale reads in amperes. This torque is opposed by a spring which exerts a torque τ_s approximately proportional to the angle through which it is turned.

Lorentz Force and velocity selector [Serway pg 839-840]

A charge moving with a velocity $\vec{v}$ in the presence of both an electric field $\vec{E}$ and a magnetic field $\vec{B}$ experience both an electric force and a magnetic force. The total (also known as the Lorentz) force is given by

$$\vec{F} = q\vec{E} + q\vec{v} \times \vec{B}$$

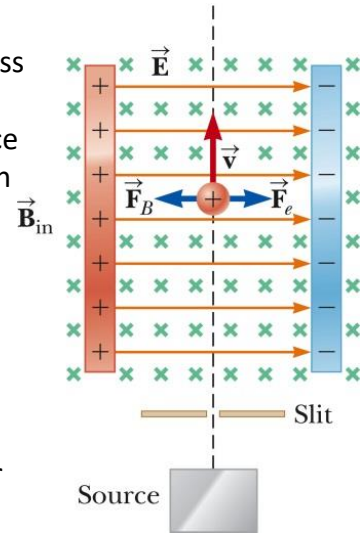
We note here that the magnetic force is velocity dependent and we can use this fact to construct a velocity selector.

In the diagram on the right, the charge q moves through a cross field (B and E field). The net force acting on it is zero and it continues to move in a straight line with constant speed. Since the velocity of the charge is perpendicular to the field, we can write

$$0 = qE - qvB$$

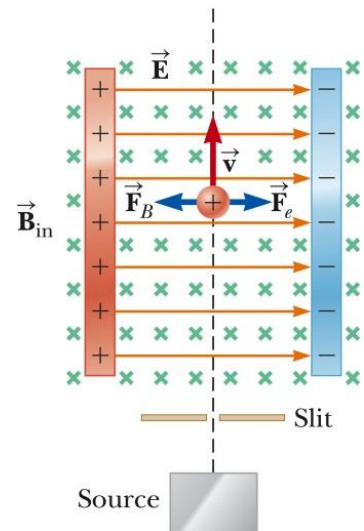
$$v = \frac{E}{B}$$

Charged particles moving faster than v will deflect to the _____ and those moving slower than v will deflect to the _____. The velocity selector is used together with mass spectrometer to separate or calculate ions of different charge-mass ratio (see earlier).



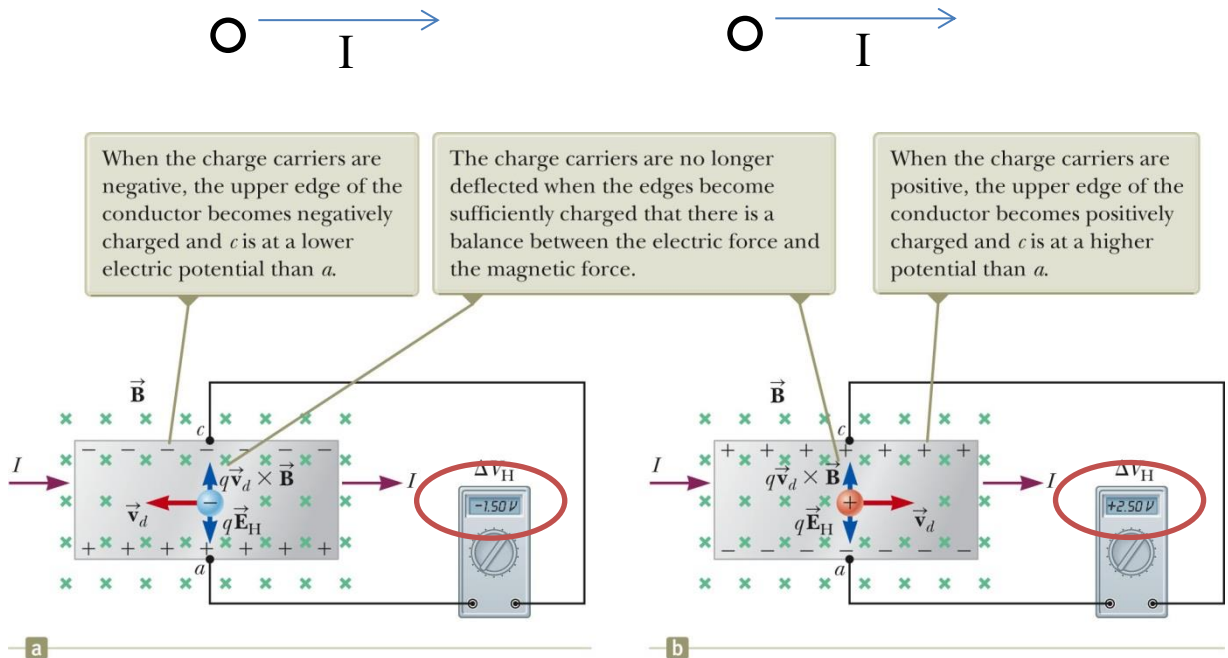
Example [Giancoli Prob 27.18 (modified)]

What is the velocity of a beam of electrons that goes undeflected when passing through perpendicular electric and magnetic fields of magnitude $8.8 \times 10^3 \text{ V/m}$ and $7.5 \times 10^{-3} \text{ T}$ respectively?



Hall Effect [Serway pg 847-849; Chap 29.6; Giancoli pg 273] Video after Exam

Is there a simple way to determine the sign of charge carriers? Consider what happens when a uniform magnetic field B is applied perpendicular to a slab of conductor with a constant current I (for charge carriers of different signs) ...



1. In the diagram, the charge carriers are first deflected to the upper edge (direction of current is fixed). This deflection causes a build up of charges on both the upper and lower edges. Hence an electric field $\vec{E}$, that causes an electric force to counter the magnetic deflection force is produced.
2. At steady state, when the magnetic deflection force balances the electric force, the charge carriers move straight and are no longer deflected. Now we can analyze using velocity selector equation:

$$qE_H = qv_d B$$

Here, E_H is the Hall field and v_d is the drift velocity.

3. If d is the width of the conductor, the Hall voltage is

$$\Delta V_H = E_H d = v_d B d.$$

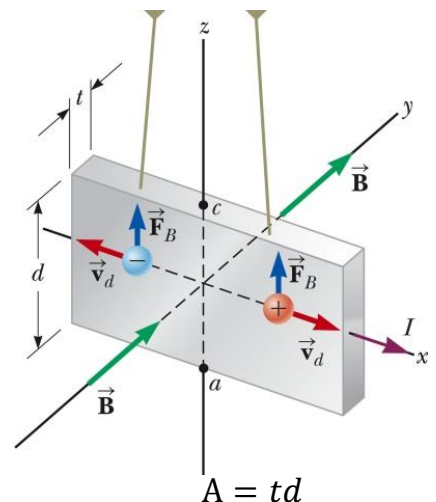
Knowing width d , Hall voltage ΔV and B , we can obtain the drift velocity v_d ! The sign of ΔV_H gives us the sign of the charges carriers.

4. We learnt earlier that $I = nqv_d A$, thus we can write

$$\Delta V_H = \frac{IBd}{nqA} = \frac{IB}{nqt}.$$

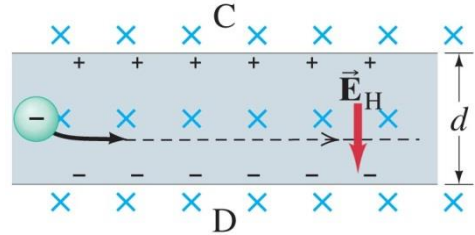
All quantities here can be easily measured to determine the Hall coefficient

$R_H = \frac{1}{nq}$. A properly calibrated conductor can be used to measure B-fields.



Giancoli pg 724 Example 27.13 Video after Exam

A long copper strip 1.8 cm wide and 1.0 mm thick is placed in a 1.2-T magnetic field. When a steady current of 15 A passes through it, the Hall emf is measured to be $1.02 \mu\text{V}$. Determine the drift velocity of the electrons and the density of free (conducting) electrons (number per unit volume) in the copper.



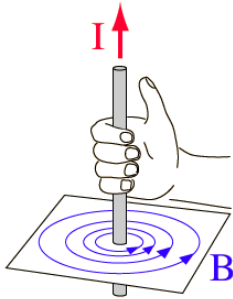
$$v_d = \frac{\mathcal{E}_H}{Bd} = \frac{1.02 \times 10^{-6} \text{V}}{(1.2 \text{T})(1.8 \times 10^{-2} \text{m})} = 4.7 \times 10^{-5} \text{m/s}$$

Using, $I = nev_d A$

$$\begin{aligned} n &= \frac{I}{ev_d A} \\ &= \frac{15 \text{ A}}{(1.6 \times 10^{-19} \text{C})(4.7 \times 10^{-5} \text{m/s})(1.8 \times 10^{-2} \text{m})(1.0 \times 10^{-3} \text{m})} \\ &= 11 \times 10^{28} \text{ m}^{-3} \end{aligned}$$

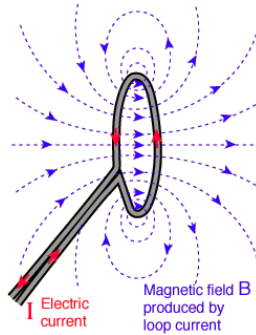
Three main ideas in Electromagnetism

1. Whenever there is a current / moving charge, there will always be a magnetic field associated with it. [Oersted 1820, Biot-Savart, Ampere's Law]



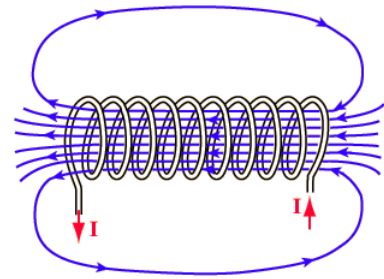
Long Wire

$$B = \frac{\mu I}{2\pi r}$$



Flat Coil

$$B_{\text{centre}} = \frac{\mu NI}{2a}$$

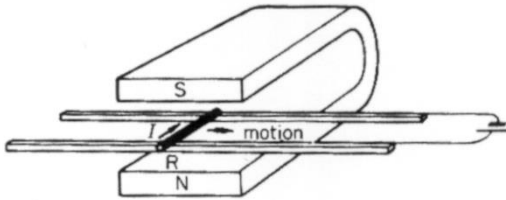


Long Solenoid

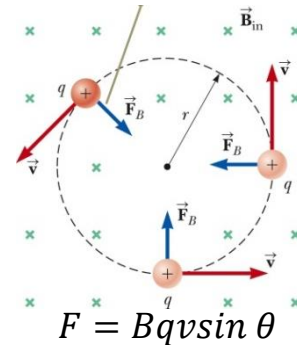
$$B_{\text{centre}} = \mu n I$$

From <http://hyperphysics.phy-astr.gsu.edu/>

2. Current / moving charge in a magnetic field produces a deflecting force. [Faraday, first electric motor, 1821; Ampere, two wires with electric current attract. 1822]



$$F = BIL \sin \theta$$



$$F = Bqv \sin \theta$$

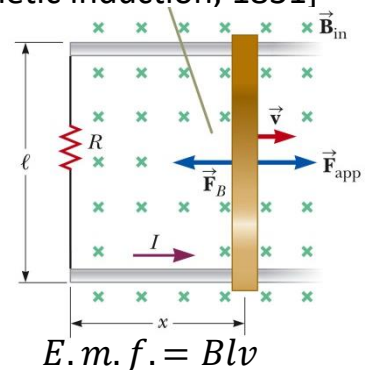
[Serway pg 836, Fig 29.8]

3. Change of magnetic field near conductor produces an induced E.M.F (Relative motion between a magnetic field and conductor produces an induced E.M.F) [Faraday's law of electromagnetic induction, 1831]



$$E.m.f. = -N \frac{d\Phi_B}{dt}; \Phi_B = BA \text{ (simplest case)}$$

$$\text{Generally, } \Phi_B = \int \vec{B} \cdot d\vec{A}$$



$$E.m.f. = Blv$$

[Serway pg 898, Fig 31.8]

Can you produce a magnetic field without a magnet?

Dr Ho Shen Yong

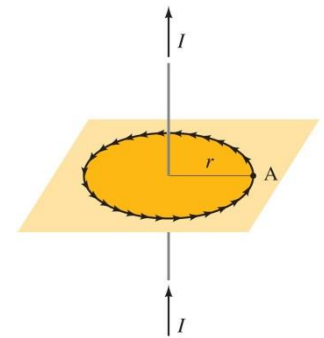
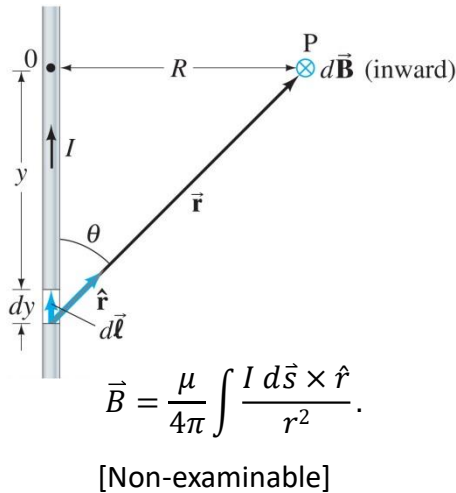
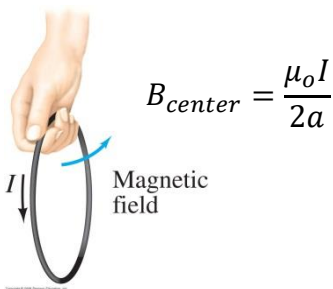
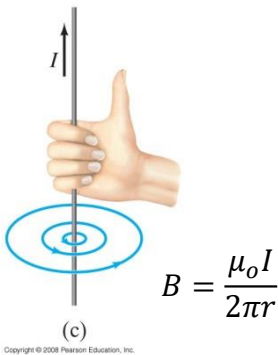
Lecturer, School of Physical and Mathematical Sciences
Nanyang Technological University

Week 10

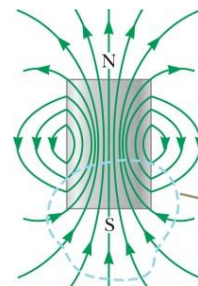
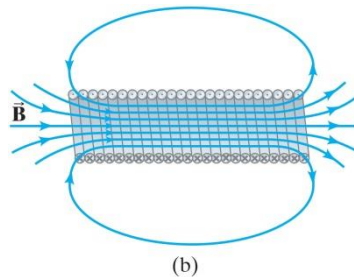
Serway Chap 29

"Tesla has contributed more to electrical science than any man up to his time."

- Statement of 1896, Lord William Thomson Kelvin (1824-1907)

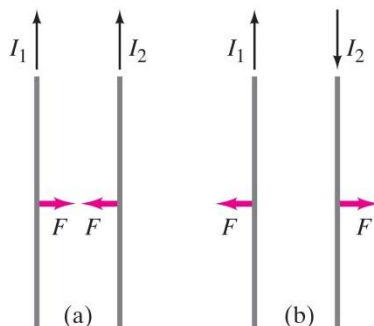


$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{net, encl}$$

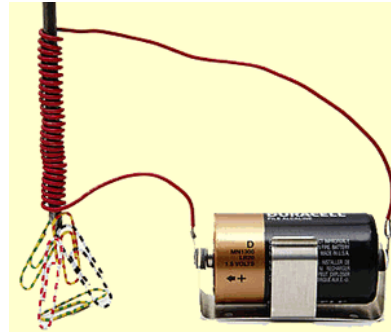
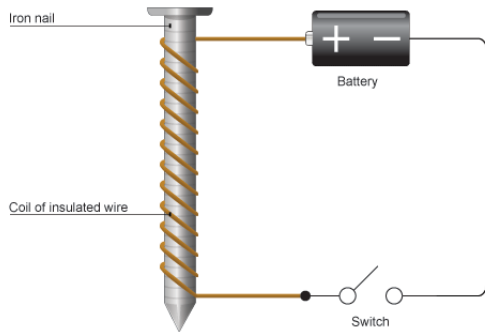


The net magnetic flux through a closed surface surrounding one of the poles or any other closed surface is zero.

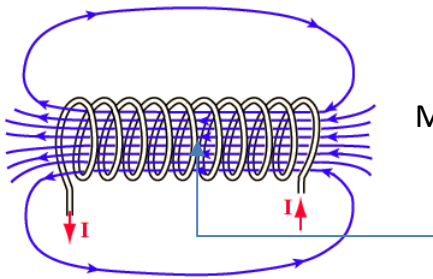
$$\oint \vec{B} \cdot d\vec{A} = 0$$



Idea 2: Current / Moving charges produces Magnetic Field



<http://www.dynamicscience.com.au/tester/solutions1/magicofsci/electromagnetinv.html>



Long Solenoid

Magnetic field in the center of a long solenoid,

$$B = \mu n I$$

where μ is the permeability of the medium in the solenoid, n is turns per unit length and I is the current. The magnetic field at the ends is approximate the magnetic field in the center

$$B_{end} \approx \frac{1}{2} B_{center}$$

Permeability / Ferromagnetic / Paramagnetic / Diamagnetic

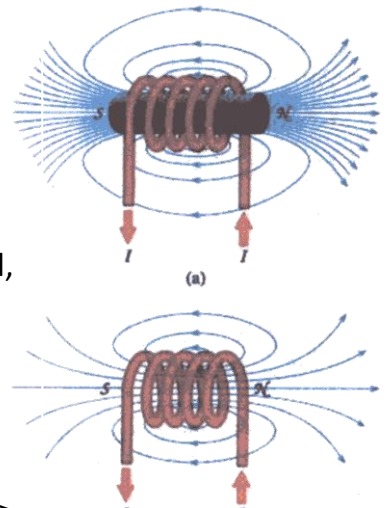
The permeability of free space is

$$\mu_0 = 4\pi \times 10^{-7} T \frac{m}{A}$$

It is well known that the magnetic field of a solenoid can be increased by inserting a piece of soft iron:

$$B = \mu_0 n I \rightarrow B' = \mu n I$$

In this case, $\mu \gg \mu_0$. When placed inside a magnetic field, the magnetic domains in the soft iron strengthen the already present magnetic field. Materials that increases the magnetic fields in this manner are described as ferromagnetic. Examples include soft iron, steel, cobalt and nickel. There are also substances that contribute slightly to an external magnetic fields (paramagnetic), $\mu \gtrsim \mu_0$ and there are some which even expel external magnetic fields (diamagnetic), $\mu < \mu_0$.



[Hecht, pg 802; Fig 21.22]

Exercise (Prob G28.25): A 32-cm-long solenoid, 1.8 cm in diameter, is to produce a 0.30-T magnetic field at its center. If the maximum current is 4.5 A, how many turns must the solenoid have? How will the magnetic field change if an iron rod is placed in the through the solenoid?

Giancoli pg 734 Magnetic Field due to a long straight wire

The magnetic field B due to a straight wire carrying current I is inversely proportional to the distance r from the wire:

$$B = \frac{\mu_0 I}{2\pi r}$$

The permeability of free space μ_0 has the value

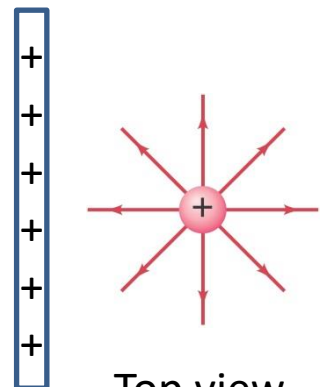
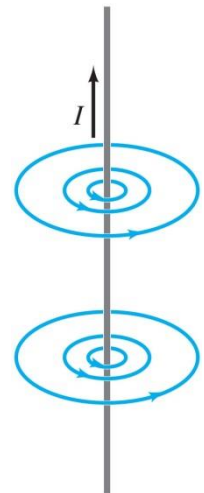
$$\mu_0 = 4\pi \times 10^{-7} \text{ T}\cdot\text{m/A}.$$

We compare it with the electric field of a long charged rod (wire with excess static charges of one sign).

The electric field E due to a straight wire carrying charge per unit length λ is inversely proportional to the distance r from the wire:

$$E = \frac{\lambda}{2\pi\epsilon_0 r} \text{ where } \lambda = \frac{Q}{l}$$

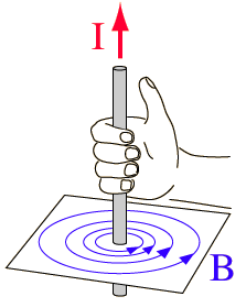
We note that the direction of the magnetic field is at right angles to both the current and the radius while the electric field is points along the radial direction.



Top view

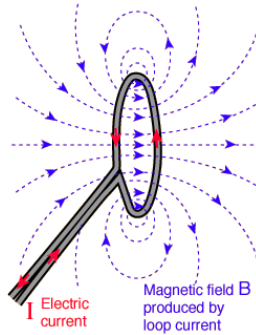
Magnetic Field due to currents of different “shapes”

Whenever there is a current / moving charge, there will always be a magnetic field associated with it. [Oersted 1820, Biot-Savart, Ampere’s Law]



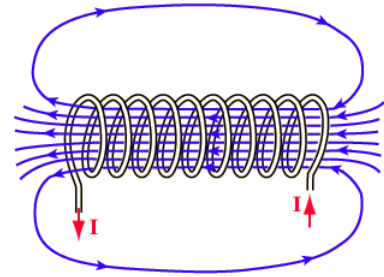
Long Wire

$$B = \frac{\mu I}{2\pi r}$$



Flat Coil

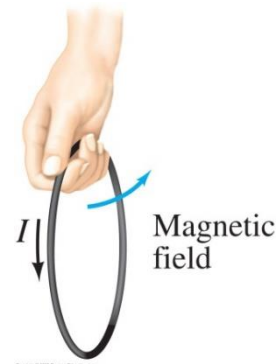
$$B_{\text{centre}} = \frac{\mu N I}{2a}$$



Long Solenoid

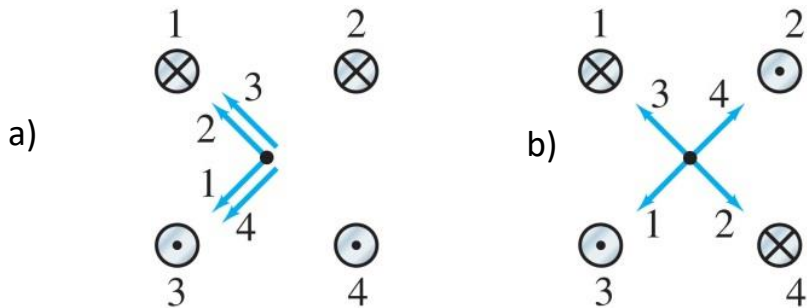
$$B_{\text{centre}} = \mu n I$$

Exercise (Prob G28.1): Jumper cables used to start a stalled vehicle often carry a 65-A current. How strong is the magnetic field 3.5 cm from one cable? Compare to the Earth’s magnetic field. ($5.0 \times 10^{-5} \text{ T}$)

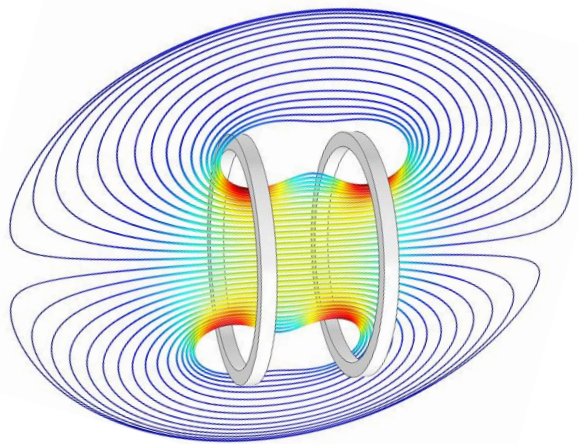


Giancoli pg 734 Example 28.2: Magnetic field is a Vector

The figures show four long parallel wires which carry equal currents into or out of the page. In which configuration, (a) or (b), is the magnetic field greater at the center of the square?

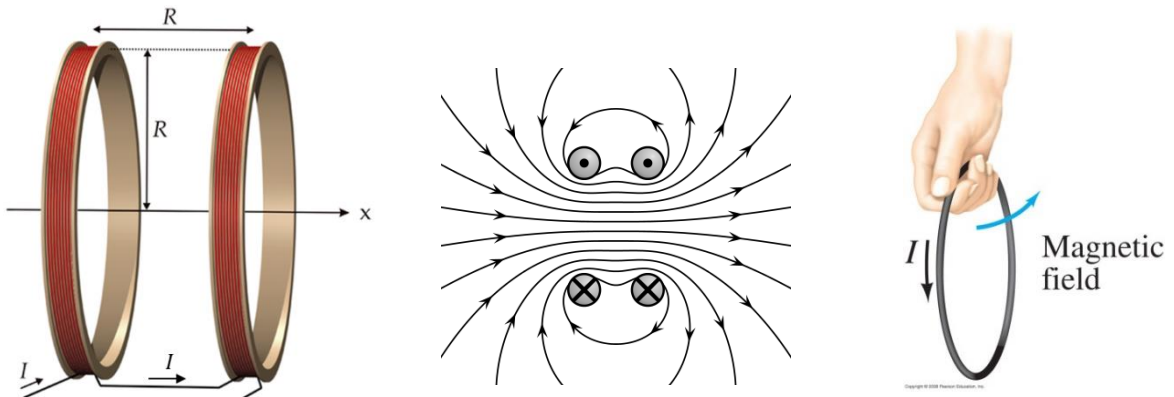


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Magnetic Field due to currents of different “shapes” (Part 2) – Helmholtz Coil

It is possible to have wires arranged in other configurations to generate the required magnetic fields. The example below is known as the Helmholtz Coil.



The magnetic field B at the mid-point between the two coils when a current I is passed is given by

$$B = \left(\frac{4}{5}\right)^{\frac{3}{2}} \frac{\mu_0 n I}{R}$$

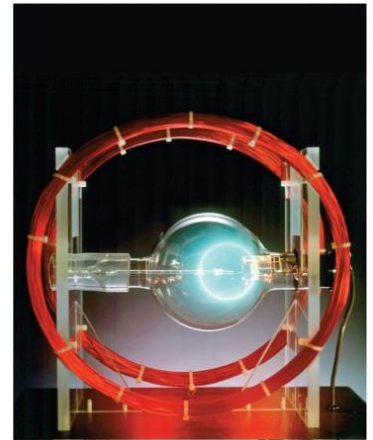
where n is the number of turns in each coil and μ_0 is the permeability of free space.

Recall the earlier question on page 8 of the lecture notes

“Giancoli pg 715, Example 27-7 :An electron travels at 2.0×10^7 m/s in a plane perpendicular to a uniform 0.010-T magnetic field. Describe its path quantitatively.”

We note that there was no magnet but there is a magnetic field of $B = 0.010\text{T}$.

- If $n = 250$ and $R = 0.250\text{ m}$, calculate the current I in the Helmholtz coil need to produce 0.010T.
- Derive an expression relating the radius r of the electron beam and the current I in the Helmholtz coil, n and R .



$$\begin{aligned} \text{a.} \quad B &= \left(\frac{4}{5}\right)^{\frac{3}{2}} \frac{\mu_0 n I}{R} \\ 0.010\text{ T} &= \left(\frac{4}{5}\right)^{\frac{3}{2}} \frac{(4 \times \pi \times 10^{-7}) 250 I}{0.250\text{ m}} \Rightarrow I = 11.1\text{ A} \end{aligned}$$

$$\begin{aligned} \text{b.} \quad B q v &= \frac{m v^2}{r} \\ \left(\frac{4}{5}\right)^{\frac{3}{2}} \frac{\mu_0 n I}{R_{\text{coil}}} q v &= \frac{m v^2}{r} \Rightarrow r = \frac{m v}{q \left(\frac{4}{5}\right)^{\frac{3}{2}} \frac{\mu_0 n I}{R_{\text{coil}}}} \end{aligned}$$

Idea 2 + Idea 1: Giancoli pg 735 Force between two parallel wires

The magnetic field produced at the position of wire 2 due to the current in wire 1 (distance d) away is

$$B_1 = \frac{\mu_o}{2\pi} \frac{I_1}{d}$$

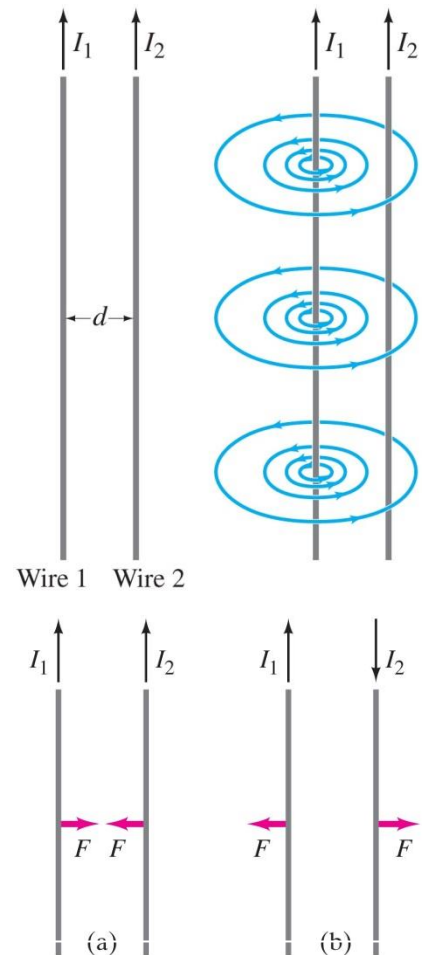
The force this field (due to wire 1) exerts on a length l of wire 2 is

$$F_2 = B_1 I_2 l = \frac{\mu_o}{2\pi} \frac{I_1 I_2}{d} l$$

Similarly, The force this field (due to wire 2) exerts on a length l of wire 1 is

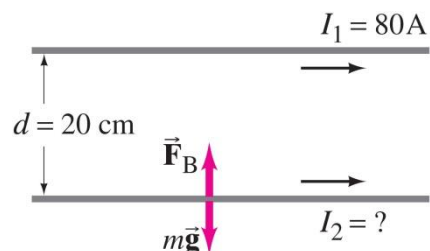
$$F_1 = B_2 I_1 l = \frac{\mu_o}{2\pi} \frac{I_2 I_1}{d} l$$

The two forces F_1 and F_2 are an action reaction pair.



Giancoli pg 736 Example 28.5

A horizontal wire carries a current $I_1 = 80$ A dc. A second parallel wire 20 cm below it must carry how much current I_2 so that it doesn't fall due to gravity? The lower wire has a mass of 0.12 g per meter of length.



Conventions and Mnemonics

Figure 1: Current and Magnetic Field

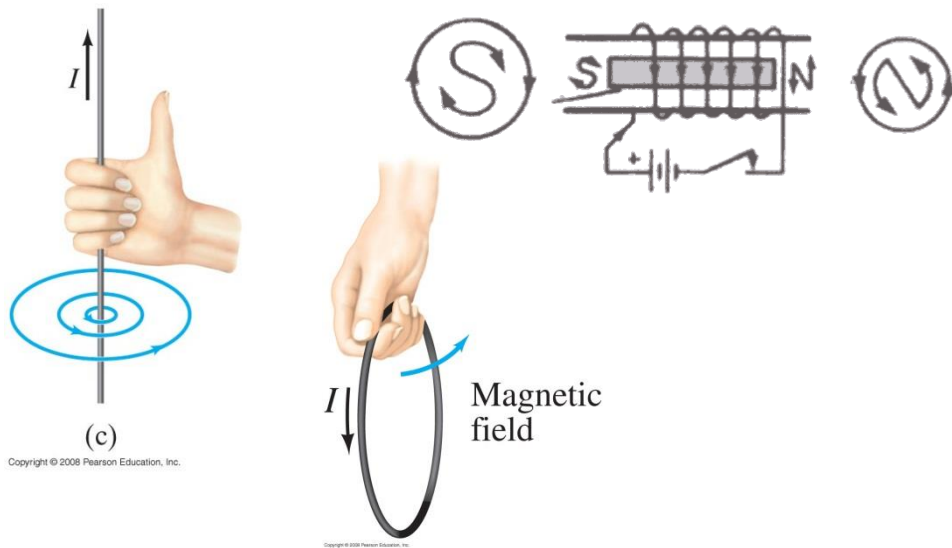


Figure 2: Current in Magnetic Field

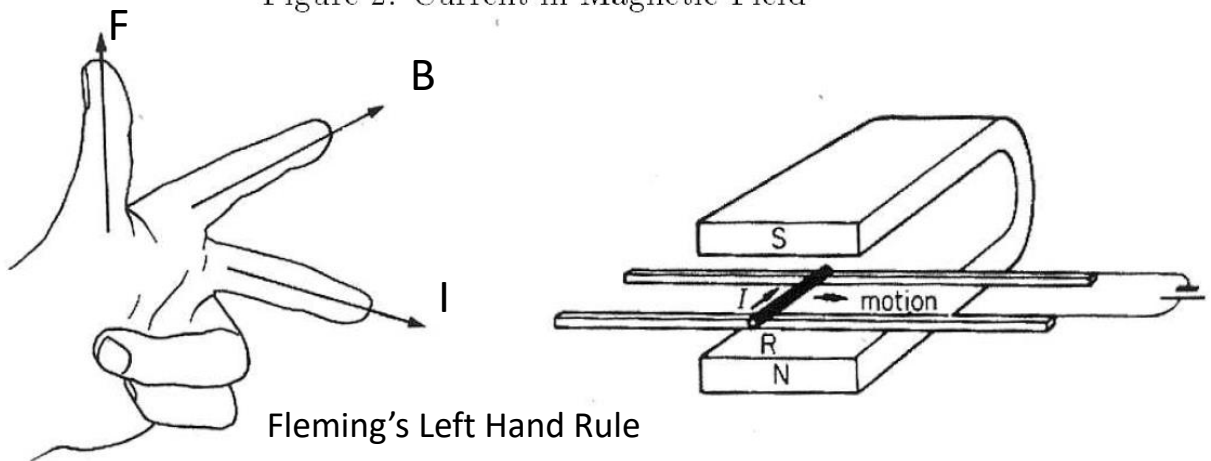


Figure 3: Induced E.M.F

